

KUSHAGRA SRIVASTAVA

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WORK EXPERIENCE

Tata Consultancy Services (TCS), India

Jan 2024 – Present

System Engineer — Backend Platform

- **Owned and scaled authentication and authorization platforms** across **100+ internal applications** and **50+ microservices**, handling **500K+ daily requests** and reducing login latency from **~5s to under 1s** through **Redis-backed caching and refined session management**.
- **Designed and executed a centralized OTP platform** processing **100K+ verifications per day**, enforcing Redis-based lifecycle management, rate limiting, configurable expiry and retry policies, and SMS fallback support without redeployments.
- Implemented and standardized **OAuth 2.0 and OpenID Connect authentication flows** (password grant, client credentials, refresh tokens) to secure access for internal and external client applications.
- Built and enforced a **centralized, database-driven RBAC framework** enabling runtime permission evaluation and centralized API registration, significantly reducing deployment cycles for access control changes.
- Architected a **cache-first session and token validation layer** using Redis, reducing database load by **50%** and enhancing responsiveness under high concurrency.
- Extended **Keycloak via custom SPIs** to employ dynamic role mappings and group-based password policies, cutting IAM rollout and configuration effort by **50%+**.
- Designed and maintained a **Kafka-based event processing pipeline** with idempotent consumers and retry handling, reliably supporting peak loads of **50K+ events per hour**.
- Employed **resilience patterns** (circuit breakers, retries, fallbacks) across backend services, maintaining **99.5% uptime** during fault-injection and failure simulations.
- Built a reusable **AWS S3 abstraction framework**, reducing storage-related code duplication by **40%** and standardizing access patterns across backend services.
- Optimized **JPA queries, indexing strategies, and connection pooling**, enhancing API response times by **20%** in performance benchmarks.
- Standardized **global exception handling and error contracts** across **200+ microservices**, reducing production debugging and incident triage time by **40%**.
- Led design reviews and mentored **10+ engineers**, driving backend best practices across authentication-critical systems.
- Designed a **dynamic JPA Specification-based query framework** for multi-criteria REST filtering with optimized joins and projections, improving complex search performance by **25%+**.
- Designed a **Redis-backed, cache-first IAM effective-permission API** that unified role–privilege–policy–permission resolution into a single call, reduced access staleness from **5+ minutes** to near real-time, and eliminated cross-service authorization round-trips.

Ksolves India, Noida, India

Dec 2020 – Jan 2024

Software Developer

- Enhanced Android application stability by **25%** through performance tuning, analytics-driven debugging, and memory optimization using Java and Kotlin.
- Executed Stripe wireless payment integration and developed a compression pipeline, reducing server storage usage by **50%**.
- Implemented **MVVM and MVP architectural patterns** to streamline application structure and improve long-term maintainability.
- Adopted Agile methodologies to enhance sprint predictability and delivery timelines.

TECHNICAL SKILLS

Java, Kotlin, Spring Boot, REST APIs, Microservices, Apache Kafka, Event-Driven Architecture, OAuth 2.0, OpenID Connect, JWT, Keycloak, Spring Security, PostgreSQL, MySQL, MongoDB, Redis, Docker, Kubernetes, AWS (EC2, S3, IAM), Jenkins, GitLab CI/CD, JUnit, Mockito, SonarQube, React

PROJECTS

RateLimitX — Multi-Tenant Rate Limiting SaaS

github.com/obj-this-self-me/rate-limiter

- Built a **Redis-backed scalable rate-limiting platform** using Token Bucket, Sliding Window, and Fixed Window algorithms, supporting **10K+ requests/sec per tenant** with **sub-5ms evaluation latency**.

EDUCATION

Bachelor of Technology, Computer Science

2016 – 2020

United College of Engineering and Research

CGPA: 7.4

CERTIFICATION

AWS Certified Developer Associate - <https://bit.ly/4rFEilr>

February, 2026 – February, 2029